

$$\frac{L-18}{ch-7.2}$$

Newton's Cote Method has $\rightarrow$ accuracy issue.
to make it better $\rightarrow$ Composite Newton's Method.

Composite Newton's Cote Method

$f(x)$ and interval $[a, b]$

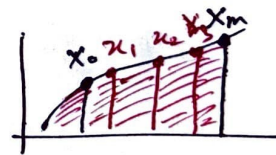
$\hookrightarrow$ will break this interval into m nodes/subintervals

The distance will be equal h distance.

Composite Newton's Cote formula of Integration

$$C_{1,m} = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + \dots + f(x_m)]$$

Ex. $f(x) = e^x$; interval $[0, 2]$,
 $m = 4$



Ans. Distance, $h = \frac{b-a}{m} = \frac{2-0}{4} = 0.5$

$$\begin{cases} x_0 = a = 0 \\ x_1 = x_0 + h = 0 + 0.5 = 0.5 \\ x_2 = x_1 + h = 0.5 + 0.5 = 1 \\ x_3 = x_2 + h = 1.5 \\ x_4 = x_3 + h = 2 \end{cases}$$

$m+1$ nodes.

$$\begin{aligned} I_{f(x)} &= \int_a^b f(x) dx = C_{1,4} = \frac{h}{2} [f(x_0) + 2f(x_1) + 2f(x_2) + 2f(x_3) + f(x_4)] \\ &= \frac{0.5}{2} [e^0 + 2 \cdot e^{0.5} + 2e^1 + 2e^{1.5} + 2e^2] \\ &= \text{ANS.} \end{aligned}$$

Note!

if $m \uparrow$ then $h \downarrow$.

→ then the integration gives more accurate result.

□ Show sub-interval h ; error goes down by factor 4. (Having quadratic convergence $O(h^2)$.)

→ such that, sub-interval $[x_{i-1}, x_i]$.

$$\text{upper bound error} \rightarrow |I(f) - I_i(f)| \leq \underbrace{\left| \frac{f''(\xi)}{2!} \right|_{\max_{\xi \in [x_{i-1}, x_i]}}}_{\xi \text{ part}} \cdot \underbrace{\int_{x_{i-1}}^{x_i} |(x-x_{i-1})(x-x_i)| dx}_{w(x) \text{ part.}}$$

$$\begin{aligned} \text{w(x) part} \rightarrow \int_{x_{i-1}}^{x_i} |(x-x_{i-1})(x-x_i)| dx &= \int_{x_{i-1}}^{x_i} [-x^2 + (x_{i-1} + x_i)x - x_{i-1}x_i] dx \\ &= \left[-\frac{x^3}{3} + \frac{1}{2}(x_{i-1} + x_i)x^2 - x_{i-1}x_ix \right]_{x_{i-1}}^{x_i} \\ &= \frac{x_i^3}{6} - \frac{1}{2}x_{i-1}x_i^2 + \frac{1}{2}x_{i-1}^2x_i - \frac{1}{6}x_{i-1}^3 \\ &= \frac{1}{6}(x_i - x_{i-1})^3 = \frac{1}{6}h^3 \end{aligned}$$

So, [plugging w(x) part]

$$|I(f) - C_{1,m}(f)| \leq \left| \frac{f''(\xi)}{2!} \right|_{\max_{\xi \in [x_{i-1}, x_i]}} \cdot \frac{mh^3}{6}$$

$$= \frac{b-a}{12} \underbrace{(h^2)}_{\text{circled}} \left| f''(\xi) \right|_{\max_{\xi \in [x_{i-1}, x_i]}}$$

→ So, convergence will be $O(h^2)$